HU Collection
ESAT Mock Set F: Mathematics 2

Worked Solutions

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Answer key

Q	1	2	3	4	5	6	7	8	9	10
Ans	B	D	A	D	A	D	C	F	B	F

Q	11	12	13	14	15	16	17	18	19	20
Ans	B	E	D	E	A	B	E	F	C	F

Q	21	22	23	24	25	26	27			
Ans	C	B	E	A	C	C	A			

Each solution gives an intended route and names the trap the question was built around. Diagrams show the configuration each solution works with.

Question 1. Answer **B**.

The rate of rise is $\dfrac{d\theta}{dt} = 27 - 2kt + 3t^2$, and both rules must hold for every t with $0 \le t \le 2$, not for every real t.

The second rule says $27 - 2kt + 3t^2 \le 27$, that is $t(3t - 2k) \le 0$, so $2k \ge 3t$ for every t up to 2. This means $k \ge 3$.

The first rule says $27 - 2kt + 3t^2 \ge 0$. At $t = 0$ this is automatic, and for $0 < t \le 2$ it means $2k \le 3t + \dfrac{27}{t}$. The function $3t + \dfrac{27}{t}$ has derivative $3 - \dfrac{27}{t^2}$, which is negative for $0 < t < 3$, so on $0 < t \le 2$ it is decreasing and its least value is at $t = 2$, namely $6 + \dfrac{27}{2} = \dfrac{39}{2}$. Hence $k \le \dfrac{39}{4}$, and the answer is $3 \le k \le \dfrac{39}{4}$. (At $k = \frac{39}{4}$ the temperature is momentarily level at the very end of the test, which the rule allows.)

Asking for $\dfrac{d\theta}{dt} \ge 0$ for all real t (discriminant at most 0) gives $-9 \le k \le 9$, and using the stationary point $t = 3$ of $3t + \dfrac{27}{t}$, which lies outside the test, gives the upper limit 9. Checking the second rule only at $t = 1$ gives $k \ge \frac{3}{2}$, and checking the first only at $t = 1$ gives $k \le 15$.

Question 2. Answer **D**.

Put $t = \sqrt{x - 4} \ge 0$, so that $x = t^2 + 4$. Then $x \pm 4t = t^2 \pm 4t + 4 = (t \pm 2)^2$, and so

$$f(x) = (t + 2) + |t - 2|.$$

The modulus is the whole question. For $0 \le t \le 2$, that is $4 \le x \le 8$, the output is $f(x) = 4$; for $t \ge 2$, that is $x \ge 8$, it is $f(x) = 2t = 2\sqrt{x - 4}$.

So the five readings $4, 5, 6, 7, 8$ are all accepted. For $x \ge 8$ we need $2\sqrt{x - 4}$ to be an integer, so $4(x - 4) = m^2$ for an integer m; then m is even, $m = 2j$, and $x - 4 = j^2$ is a perfect square. This gives $x = 4 + j^2$ for $j = 2, 3, \ldots, 9$, that is $8, 13, 20, \ldots, 85$, and $x = 8$ has already been counted. The total is $5 + 7 = 12$.

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Writing $\sqrt{(t-2)^2} = t-2$ for every t gives $f = 2t$ throughout and the answer 10; writing it as $2-t$ throughout gives $f = 4$ for every reading and the answer 97; counting $x = 8$ twice gives 13.

Question 3. Answer **A**.

Integrating the deceleration, with initial speed u,

$$v = u - \tfrac{2}{3}kt^{3/2}, \qquad s = ut - \tfrac{4}{15}kt^{5/2}.$$

She stops at $t = 4$, so $u = \tfrac{2}{3}k \cdot 8 = \tfrac{16k}{3}$, that is $k = \tfrac{3u}{16}$. Then

$$24 = s(4) = 4u - \tfrac{4}{15}k \cdot 32 = 4u - \tfrac{128}{15} \cdot \tfrac{3u}{16} = 4u - \tfrac{8u}{5} = \tfrac{12u}{5},$$

so $u = 10$ m s^{-1}.

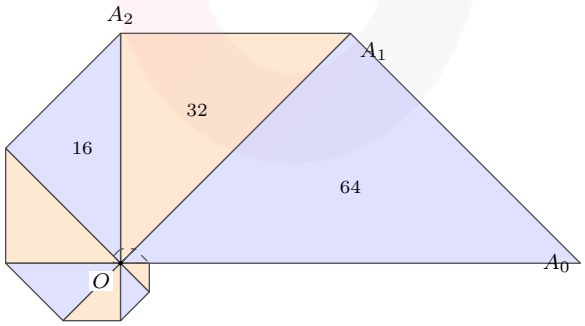
Assuming a constant deceleration gives $s = \tfrac{1}{2}uT$ and $u = 12$; a deceleration proportional to t gives $s = \tfrac{2}{3}uT$ and $u = 9$. Keeping only the term $\tfrac{4}{15}kt^{5/2}$ for the distance gives 15, and 6 is just the average speed.

Question 4. Answer **D**.

Each triangle has an angle of 45° at O, and its shorter sides are $\tfrac{1}{\sqrt{2}}$ times its hypotenuse. So each triangle is the previous one rotated through 45° about O and scaled by $\tfrac{1}{\sqrt{2}}$, and its area is half the previous area. The first triangle has shorter sides $8\sqrt{2}$ and area 64, so the areas are $64, 32, 16, \ldots$, with sum 128.

But 128 counts some points more than once. After eight triangles the pattern has turned through 360°, so triangle 9 is triangle 1 scaled by $\left(\tfrac{1}{\sqrt{2}}\right)^8 = \tfrac{1}{16}$ about O. Since O is a vertex of triangle 1, this copy lies inside triangle 1. In the same way every later triangle lies inside one of the first eight, and the first eight fill the eight different 45° angles at O without overlapping. The shaded area is therefore

$$64\left(1 + \tfrac{1}{2} + \tfrac{1}{4} + \cdots + \tfrac{1}{2^7}\right) = 128\left(1 - \tfrac{1}{256}\right) = \tfrac{255}{2}.$$



Stopping after seven or after four triangles (thinking each turns through 90°) gives 127 or 120, and scaling areas by $\tfrac{1}{4}$ instead of $\tfrac{1}{2}$ gives $\tfrac{256}{3}$.

Question 5. Answer **A**.

With the extra measurement at $x = c$ the estimate is

$$T(c) = \frac{c}{2}\left(0 + \sqrt{c}\right) + \frac{9-c}{2}\left(\sqrt{c} + 3\right) = \frac{9\sqrt{c} + 27 - 3c}{2}.$$

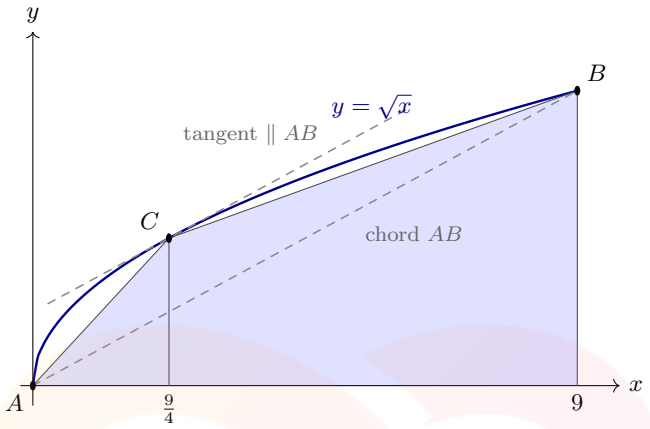
Since the curve $y = \sqrt{x}$ bends downwards, both trapezia lie under it, so every estimate is too small and the most accurate one is the largest.

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$T'(c) = \dfrac{9}{4\sqrt{c}} - \dfrac{3}{2} = 0$ gives $\sqrt{c} = \dfrac{3}{2}$, so $c = \dfrac{9}{4}$, and

$$T\left(\tfrac{9}{4}\right) = \frac{1}{2}\left(\frac{27}{2} + 27 - \frac{27}{4}\right) = \frac{135}{8}.$$

The true area is $\displaystyle\int_0^9 \sqrt{x}\,dx = \tfrac{2}{3}\cdot 27 = 18$, so the error is $18 - \dfrac{135}{8} = \dfrac{9}{8}$.



Geometrically, the two trapezia are the single trapezium on the whole base plus the triangle ACB, so the best point C is the point of the curve furthest from the chord AB, where the tangent is parallel to AB: $\dfrac{1}{2\sqrt{c}} = \dfrac{1}{3}$. The midpoint $c = \dfrac{9}{2}$ gives an error of $\dfrac{45 - 27\sqrt{2}}{4} \approx 1.70$, the convenient choices $c = 4$ or $c = 1$ give $\dfrac{3}{2}$, a single strip gives $\dfrac{9}{2}$, and $\dfrac{9}{4}$ is the best position, not the error.

Question 6. Answer **D**.

A fall of ΔM in magnitude divides the energy by $10^{1.5\Delta M}$. With $\log_{10} 2 = 0.3$ we have $10^{0.3} = 2$. So a fall of 0.1 divides the energy by $10^{0.15} = 2^{1/2} = \sqrt{2}$, and a fall of 0.8 divides it by $10^{1.2} = 2^4 = 16$. The first aftershock therefore carries $\frac{1}{16}$ of the main shock's energy, and the aftershock energies form a geometric series with common ratio $\frac{1}{\sqrt{2}}$. Their total, as a fraction of the main shock, is

$$\frac{1}{16}\cdot\frac{1}{1-\frac{1}{\sqrt{2}}} = \frac{1}{16}\cdot\frac{\sqrt{2}}{\sqrt{2}-1} = \frac{\sqrt{2}\left(\sqrt{2}+1\right)}{16} = \frac{2+\sqrt{2}}{16}.$$

The magnitudes form an arithmetic sequence, but the energies form a geometric one. Taking ratio $\frac{1}{2}$ (as if each step were 0.2) gives $\frac{1}{8}$; starting the series at the second aftershock gives $\frac{1+\sqrt{2}}{16}$; dividing by 8 instead of 16 for the first aftershock gives $\frac{2+\sqrt{2}}{8}$.

Question 7. Answer **C**.

Write the total cost as $C(r) = Ar^{3/2} + Br^{-4}$ with positive constants A and B. Then $C'(r) = \frac{3}{2}Ar^{1/2} - 4Br^{-5}$, which is negative for small r and positive for large r, so the least cost occurs at the single stationary point, where

$$r^{11/2} = \frac{8B}{3A}.$$

Multiplying $C'(r) = 0$ by r shows that there $\frac{3}{2}Ar^{3/2} = 4Br^{-4}$, so the pumping cost is $\frac{3}{8}$ of the pipe cost, and the least total cost is

$$C_{\min} = \tfrac{11}{8}Ar^{3/2} = \tfrac{11}{8}A\left(\frac{8B}{3A}\right)^{3/11},$$

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which is proportional to $B^{3/11}$.

Replacing B by $4B$ multiplies the least cost by $4^{3/11} = 2^{6/11} \approx 1.46$. The trap $2^{4/11}$ is the factor by which the best radius grows (since r is proportional to $B^{2/11}$), and $2^{16/11}$ comes from using the pipe's share $\frac{8}{11}$ of the cost as the exponent instead of the pumping share $\frac{3}{11}$.

Question 8. Answer **F**.

Since

$$\int_{t-1}^{t} s^2\,ds = \frac{t^3-(t-1)^3}{3} = t^2 - t + \frac{1}{3} \qquad \text{and} \qquad \int_{t-1}^{t} s\,ds = t - \frac{1}{2},$$

a quadratic T produces readings with the same t^2 coefficient, so $T(s) = s^2 + bs + c$, and its readings are

$$t^2 - t + \tfrac{1}{3} + b\left(t - \tfrac{1}{2}\right) + c.$$

Matching with $t^2 - 6t + 12$ gives $b = -5$ and $c = 12 - \frac{1}{3} - \frac{5}{2} = \frac{55}{6}$. So

$$T(s) = s^2 - 5s + \tfrac{55}{6} = \left(s - \tfrac{5}{2}\right)^2 + \tfrac{35}{12},$$

and the lowest actual temperature is $\frac{35}{12} \approx 2.92\,^\circ\text{C}$, at $s = 2.5$, half an hour before the lowest reading.

The averaging both delays the minimum and smooths it out: the readings never go below 3, which is the tempting answer. Getting the sign of the correction wrong gives $\frac{37}{12}$.

Question 9. Answer **B**.

For $P = (x, y)$,

$$PA^2+PB^2+PC^2 = 3\left(x^2+y^2\right)-2(0+9+6)x-2(0+0+9)y+(0+81+117) = 3\left(x^2+y^2\right)-30x-18y+198.$$

On the circle, $x^2+y^2 = 4x+6y+12$, so the load becomes $12x+18y+36-30x-18y+198 = 234-18x$. The quadratic terms and even the y terms disappear, and the load depends only on x.

The circle is $(x-2)^2 + (y-3)^2 = 25$, so x runs from -3 to 7. The greatest load is $234 + 54 = 288$, at $P = (-3, 3)$, and the least is $234 - 126 = 108$, at $(7, 3)$.

Equivalently, with $G(5, 3)$ the centroid of the transmitters, the load is $3PG^2 + 96$, and PG runs from $5 - 3 = 2$ to $5 + 3 = 8$. Forgetting the constant 96 gives 192, and 180 is the difference between the greatest and least loads.

Question 10. Answer **F**.

For a fixed wall distance X, the height of the jet at the wall is a quadratic in k:

$$y = -\frac{X^2}{20}k^2 + Xk - \frac{X^2}{20} = -\frac{X^2}{20}\left(k - \frac{10}{X}\right)^2 + 5 - \frac{X^2}{20}.$$

Without the restriction, the best choice is $k = \frac{10}{X}$, giving height $5 - \frac{X^2}{20}$. This is allowed only when $\frac{10}{X} \le 2$, that is $X \ge 5$, and then a height of at least 3 needs $X^2 \le 40$. This gives $5 \le X \le 2\sqrt{10}$.

When $X < 5$ the vertex is at some $k > 2$, so on $0 < k \le 2$ the height increases with k and the best choice is $k = 2$, giving height $2X - \frac{X^2}{4}$. A height of at least 3 needs $X^2 - 8X + 12 \le 0$, that is $2 \le X \le 6$, so here $2 \le X < 5$. Together, the answer is $2 \le X \le 2\sqrt{10}$. (A positive height at the wall means the jet has not yet landed, since the curve is above the ground between the nozzle and the landing point.)

Ignoring the mounting gives $0 < X \le 2\sqrt{10}$, but a nearby wall can only be hit high up by a steep jet, which is not allowed. Using $k = 2$ for every wall gives $2 \le X \le 6$.

Question 11. Answer **B**.

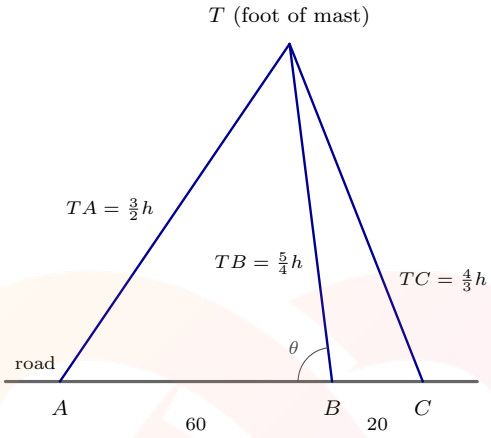
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Let the mast have height h and foot T. The horizontal distances are $TA = \frac{3}{2}h$, $TB = \frac{5}{4}h$ and $TC = \frac{4}{3}h$. Let $\theta = \angle TBA$, so that $\angle TBC = 180° - \theta$ and $\cos\angle TBC = -\cos\theta$. The cosine rule in the ground triangles TBA and TBC gives

$$TA^2 = TB^2 + 60^2 - 120\,TB\cos\theta, \qquad TC^2 = TB^2 + 20^2 + 40\,TB\cos\theta.$$

Multiplying the second equation by 3 and adding eliminates the unknown angle:

$$TA^2 + 3\,TC^2 = 4\,TB^2 + 3600 + 1200.$$



Substituting, $\left(\frac{9}{4} + \frac{16}{3} - \frac{25}{4}\right)h^2 = 4800$, that is $\frac{4}{3}h^2 = 4800$, so $h^2 = 3600$ and $h = 60$ m. Then $TA = 90$, $TB = 75$ and $TC = 80$, which do fit together: $\cos\theta = \frac{75^2+60^2-90^2}{2\cdot75\cdot60} = \frac{1}{8}$.

The point of the question is that the angle at B is unknown; it is not 90°, and B is not the midpoint of AC, so neither Pythagoras nor the midpoint form of this identity applies directly.

Question 12. Answer **E**.

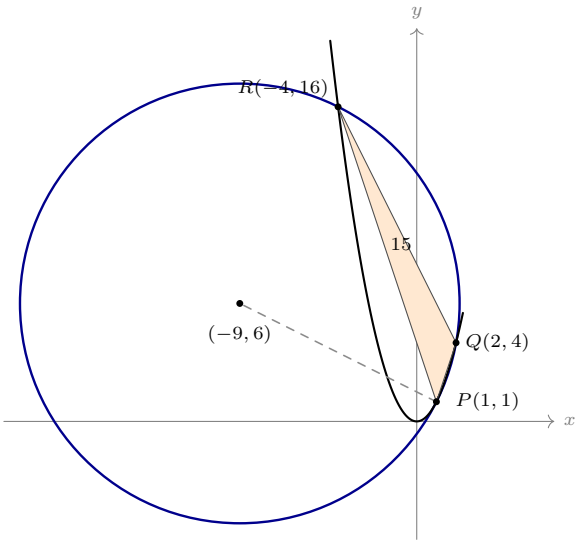
Write the circle as $x^2 + y^2 + Dx + Ey + F = 0$. Its intersections with $y = x^2$ satisfy

$$x^4 + (1 + E)x^2 + Dx + F = 0,$$

a quartic with no x^3 term, so its four roots, counted with multiplicity, add up to 0. Tangency at P makes $x = 1$ a double root, and Q gives the root 2, so the fourth root is $-(1 + 1 + 2) = -4$ and $R = (-4, 16)$.

Without this shortcut: the centre lies on the normal to the parabola at P, which has gradient $-\frac{1}{2}$, so it is $(1 - 2t,\, 1 + t)$ for some t. Equal distances to P and Q give $t = 5$, so the centre is $(-9, 6)$ and the radius is $\sqrt{125}$; then dividing the quartic by $(x - 1)^2(x - 2)$ leaves $x + 4$.

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With $\overrightarrow{PQ} = (1, 3)$ and $\overrightarrow{PR} = (-5, 15)$, the area is $\frac{1}{2}\,|1 \times 15 - 3 \times (-5)| = 15$. Forgetting that the tangency counts twice gives $R = (-3, 9)$ and area 10; counting Q twice instead gives $R = (-5, 25)$ and area 21.

Question 13. Answer **D**.

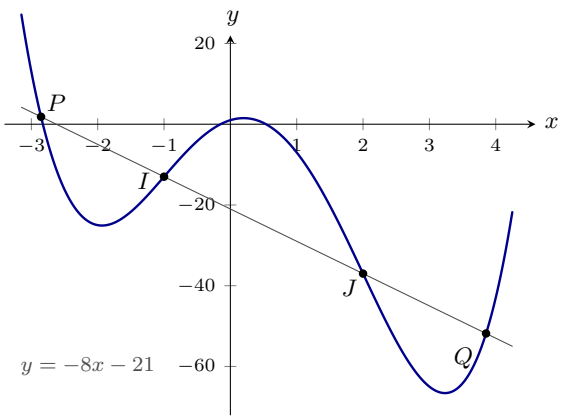
Here $\dfrac{dy}{dx} = 4x^3 - 6x^2 - 24x + 5$ and $\dfrac{d^2y}{dx^2} = 12x^2 - 12x - 24 = 12(x + 1)(x - 2)$, so the gradient is stationary at $x = -1$ and $x = 2$, giving $I(-1, -13)$ and $J(2, -37)$. The line IJ has gradient -8 and equation $y = -8x - 21$.

The curve minus the line is $x^4 - 2x^3 - 12x^2 + 13x + 22$, which must have the factors $x + 1$ and $x - 2$:

$$x^4 - 2x^3 - 12x^2 + 13x + 22 = (x^2 - x - 2)(x^2 - x - 11).$$

So P and Q have $x = \frac{1 \mp 3\sqrt{5}}{2}$. The four points are collinear, so lengths along the line are in the same ratio as differences of x-coordinates:

$$PI : IJ = \left(-1 - \tfrac{1-3\sqrt{5}}{2}\right) : 3 = \tfrac{3\sqrt{5}-3}{2} : 3 = (\sqrt{5} - 1) : 2.$$



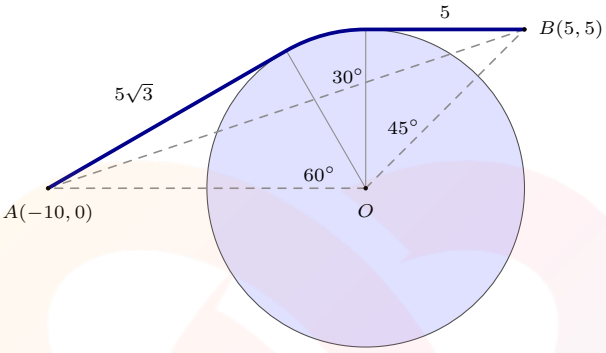
This is the golden ratio, and the same happens for every quartic curve with two such points: adding a linear function to y, rescaling y and replacing x by $px + q$ do not change the ratio, and every such quartic can be brought to $y = x^4 - 6x^2$ in this way, for which the four points are at $x = -\sqrt{5}, -1, 1, \sqrt{5}$.

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Question 14. Answer **E**.

The segment AB lies on the line $x - 3y + 10 = 0$, at distance $\frac{10}{\sqrt{10}} = \sqrt{10} < 5$ from O, so it crosses the pond and the straight route of length $5\sqrt{10}$ is not available. The shortest route is then the path of a taut string pulled round the pond: a tangent from A to the circle, an arc of the circle, and a tangent from the circle to B.

From A, with $OA = 10$, the tangent has length $\sqrt{100 - 25} = 5\sqrt{3}$, and the radius to its point of contact makes an angle of $60°$ with OA (its cosine is $\frac{5}{10}$). From B, with $OB = 5\sqrt{2}$, the tangent has length 5 and the corresponding angle is $45°$. Going round the top of the pond, the points of contact are at angles $180° - 60° = 120°$ and $45° + 45° = 90°$ from the positive x-axis, namely $\left(-\frac{5}{2}, \frac{5\sqrt{3}}{2}\right)$ and $(0, 5)$. The arc between them subtends $30° = \frac{\pi}{6}$ and has length $\frac{5\pi}{6}$.



Round the bottom, the points of contact are at $240°$ and $0°$, the arc subtends $120°$, and the route $5 + 5\sqrt{3} + \frac{10\pi}{3}$ is longer. The shortest route is $5 + 5\sqrt{3} + \frac{5\pi}{6}$ metres.

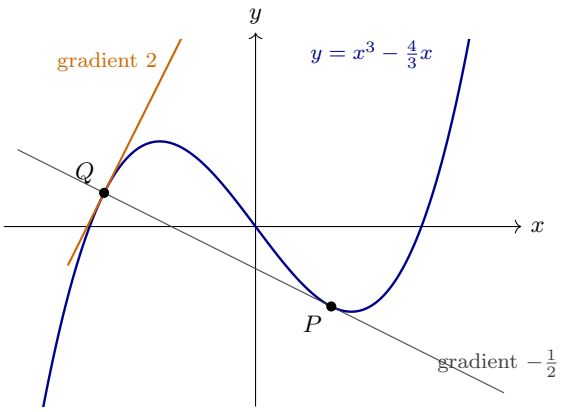
Question 15. Answer **A**.

Let P have x-coordinate $t \neq 0$ and write $f(x) = x^3 - kx$. Where the tangent at P meets the curve, $f(x)$ minus the tangent's expression vanishes. This cubic has a double root at $x = t$, and since f has no x^2 term its three roots add up to 0, so the third root is $-2t$: Q has x-coordinate $-2t$, and we never need the equation of the tangent.

The gradients there are $f'(t) = 3t^2 - k$ and $f'(-2t) = 12t^2 - k$, so with $u = t^2 > 0$ the condition is

$$(3u - k)(12u - k) = -1, \qquad \text{that is} \qquad 36u^2 - 15ku + k^2 + 1 = 0,$$

and we need a positive root u. The discriminant is $225k^2 - 144\left(k^2 + 1\right) = 81k^2 - 144$, so real roots need $|k| \geq \frac{4}{3}$. The product of the roots is $\frac{k^2+1}{36} > 0$, so they have the same sign as their sum $\frac{15k}{36}$. Hence a positive root exists exactly when $k \geq \frac{4}{3}$.



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At $k = \frac{4}{3}$ the double root is $u = \frac{5}{18}$ and the two gradients are $-\frac{1}{2}$ and 2. Forgetting that $u = t^2$ must be positive gives $k \le -\frac{4}{3}$ or $k \ge \frac{4}{3}$; but for $k < 0$ both gradients $3u - k$ and $12u - k$ are positive, so their product can never be -1.

Question 16. Answer **B**.

Working out eleven products of large binomial coefficients is not the way. Instead, recognise a coefficient. We have $(1-x)^{12} = \sum_k (-1)^k \binom{12}{k} x^k$ and $(1+x)^{12} = \sum_j \binom{12}{j} x^j$, and the coefficient of x^{10} in their product collects exactly the terms with $k + j = 10$, which is the given sum.

But $(1-x)^{12}(1+x)^{12} = \left(1-x^2\right)^{12} = \sum_m (-1)^m \binom{12}{m} x^{2m}$, and its x^{10} term comes from $m = 5$. The sum is therefore $(-1)^5 \binom{12}{5} = -792$.

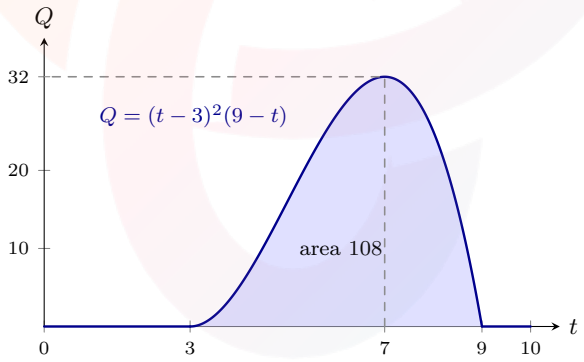
Pairing the terms for k and $10 - k$ does not make them cancel, because 10 is even and paired terms have the same sign, so 0 is wrong. The value 792 forgets the sign $(-1)^5$, and $\pm 924 = \pm\binom{12}{6}$ and $\pm 495 = \pm\binom{12}{4}$ come from the wrong power of x^2.

Question 17. Answer **E**.

Arrivals exceed 63 per hour when $30t - 3t^2 > 63$, that is $(t-3)(t-7) < 0$, so the queue starts at $t = 3$. From then on, while there is a queue, its length Q changes at the rate $r(t) - 63$, so

$$Q(t) = \int_3^t \left(30s - 3s^2 - 63\right) ds = -t^3 + 15t^2 - 63t + 81 = (t-3)^2(9-t).$$

(The factor $(t-3)^2$ is expected, since Q and Q' both vanish at $t = 3$.) The queue is longest, 32 customers, at $t = 7$, when arrivals drop back to 63 per hour, but it does not vanish then: it lasts until $t = 9$.



Translating the graph 3 units to the left, the total waiting time is

$$\int_3^9 (t-3)^2(9-t)\, dt = \int_0^6 u^2(6-u)\, du = 432 - 324 = 108 \text{ customer-hours.}$$

Stopping at $t = 7$, when the queue is longest, gives 64; 32 is the greatest length of the queue, not an area; and treating the graph as a triangle of height 32 on the base from $t = 3$ to $t = 9$ gives 96.

Question 18. Answer **F**.

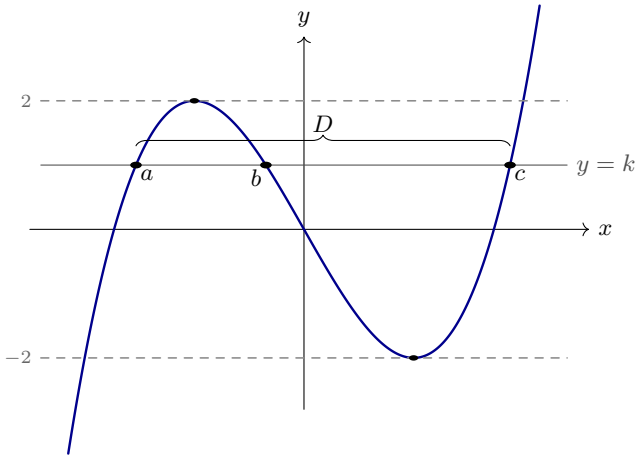
The line $y = k$ meets the curve three times exactly when k lies between the stationary values: $y' = 3x^2 - 3$ gives turning points $(-1, 2)$ and $(1, -2)$, so $-2 < k < 2$. Let the meeting points have x-coordinates $a < b < c$, the roots of $x^3 - 3x - k = 0$. Then

$$a + b + c = 0, \qquad ab + bc + ca = -3.$$

So $a + c = -b$ and $ac = -3 - b(a+c) = b^2 - 3$, giving

$$D^2 = (c-a)^2 = (a+c)^2 - 4ac = b^2 - 4\left(b^2 - 3\right) = 12 - 3b^2.$$

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The middle point lies on the falling part of the curve between the turning points, so $-1 < b < 1$, and every such b occurs (take $k = b^3 - 3b$). Hence $9 < D^2 \leq 12$, that is $3 < D \leq 2\sqrt{3}$, with the largest value at $k = 0$. The value 3 is approached as k tends to ± 2 (for example, $k = 2$ gives the points $x = -1, -1, 2$), but there two of the points merge, so $D = 3$ itself does not occur; and D is largest in the middle of the range, not near its ends.

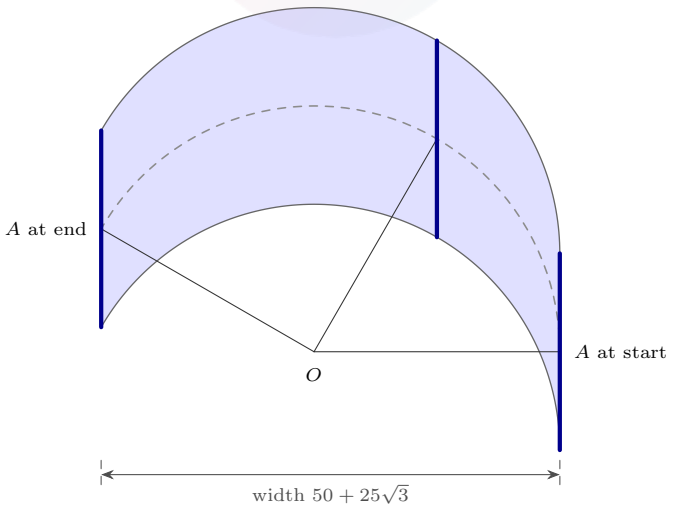
Question 19. Answer **C**.

The conventional wiper sweeps an annular sector of area

$$\tfrac{1}{2} \cdot \tfrac{5\pi}{6} \left(70^2 - 30^2\right) = \tfrac{5\pi}{12} \cdot 4000 = \tfrac{5000\pi}{3}.$$

This happens to equal the blade length times the length of the arc traced by the blade's midpoint, $40 \times 50 \cdot \frac{5\pi}{6}$.

For the pantograph wiper, put O at the origin. When the arm makes angle θ with the horizontal, the blade is the vertical segment at $x = 50\cos\theta$ running from height $50\sin\theta - 20$ to $50\sin\theta + 20$. As θ goes from 0 to $\frac{5\pi}{6}$, x decreases from 50 to $-25\sqrt{3}$ without ever turning back, so each vertical line in this range is met by exactly one position of the blade, in a segment of length 40. The wiped region is the band between the curve traced by the bottom of the blade and the same curve raised by 40, so its area is the blade length times the horizontal width: $40\left(50 + 25\sqrt{3}\right) = 2000 + 1000\sqrt{3}$.



The difference is $\frac{5000\pi}{3} - 2000 - 1000\sqrt{3} \approx 1504$. Assuming the areas are equal because the midpoints follow the same arc gives 0. Using the straight-line distance between the first and last positions of A, which is $100\sin 75^\circ = 25\left(\sqrt{6} + \sqrt{2}\right)$, in place of the horizontal width gives the third option.

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Question 20. Answer **F**.

Let $f(x) = x^3 - 6x^2 + 9x$ and let $L\left(0, \frac{27}{4}\right)$ be the light. The tangent at the point where $x = t$ is $y = f(t) + f'(t)(x - t)$, and it passes through L when $f(t) - tf'(t) = \frac{27}{4}$. Since $f(t) - tf'(t) = 6t^2 - 2t^3$, this is

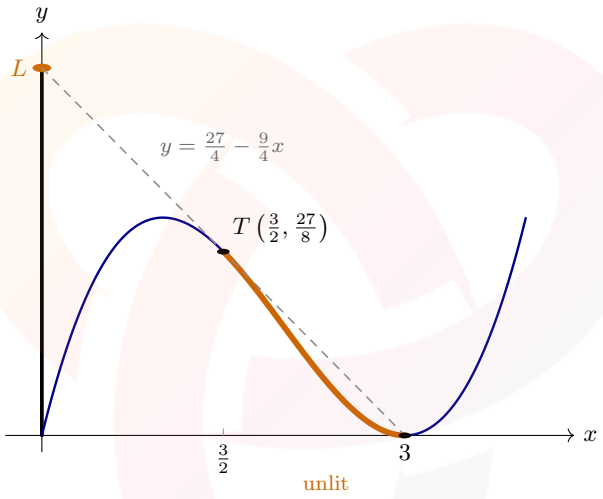
$$8t^3 - 24t^2 + 27 = 0, \qquad \text{that is} \qquad (2t - 3)\left(4t^2 - 6t - 9\right) = 0,$$

so $t = \frac{3}{2}$ or $t = \frac{3 \pm 3\sqrt{5}}{4}$.

The negative root is behind the pole. For $t = \frac{3+3\sqrt{5}}{4} \approx 2.43$ the line from L touches the curve in the hollow, but on the way it passes underneath the hill (at $x = \frac{3}{2}$ it is at height about 3.1, below the ground at 3.375), so it is not a ray of light at all. The grazing ray is the tangent at $T\left(\frac{3}{2}, \frac{27}{8}\right)$, the line $y = \frac{27}{4} - \frac{9}{4}x$, and

$$f(x) - \left(\tfrac{27}{4} - \tfrac{9}{4}x\right) = \left(x - \tfrac{3}{2}\right)^2 (x - 3).$$

So the ground lies below this ray for $\frac{3}{2} < x < 3$ and meets it again at the bottom of the hollow, $(3, 0)$; beyond $x = 3$ the ground rises above the ray and is lit again.



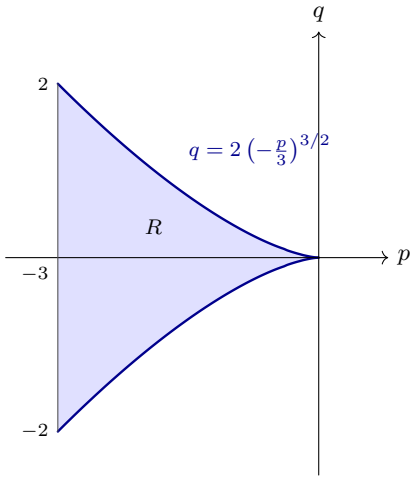
The unlit part runs from $x = \frac{3}{2}$ to $x = 3$, a horizontal extent of $\frac{3}{2}$ m. Taking the shadow to start at the summit gives 2 (or 3, for a horizontal ray from the summit to $x = 4$), and ending it at the second tangency point 2.43 gives $\frac{3\sqrt{5}-3}{4}$.

Question 21. Answer **C**.

The equation has three distinct real solutions exactly when the horizontal line $y = -q$ meets the curve $y = x^3 + px$ three times. If $p = 0$ the curve is increasing and this never happens. If $p < 0$, write $p = -3s^2$ with $s > 0$; then $y' = 3x^2 - 3s^2$, the turning points are at $x = \mp s$ with heights $\pm 2s^3$, and we need $-2s^3 < q < 2s^3$, that is

$$|q| < 2\left(-\frac{p}{3}\right)^{3/2}.$$

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So R is the region between the curves $q = \pm 2\left(-\frac{p}{3}\right)^{3/2}$ for $-3 \le p \le 0$, which meet in a cusp at the origin. Reflecting in the q-axis, which does not change areas, and using the symmetry in the p-axis,

$$\text{area} = 2\int_0^3 2\left(\frac{u}{3}\right)^{3/2} du = \frac{4}{3\sqrt{3}} \cdot \frac{2}{5} \cdot 3^{5/2} = \frac{24}{5}.$$

Counting only $q > 0$ gives $\frac{12}{5}$, dropping the $\frac{1}{3}$ inside the bracket gives $\frac{72\sqrt{3}}{5}$, and joining the corners $(-3, \pm 2)$ to the origin with straight lines gives the triangle area 6.

Question 22. Answer **B**.

Let the top row be $a,\ a+d,\ a+2d$ and let the columns have common ratios p, q, r. Write $u = a$ and $w = a + 2d$, so that $u + w = 2(a + d)$, which is nonzero because it is twice the top middle entry. The second and third rows are arithmetic, so

$$(u + w)q = up + wr, \qquad (u + w)q^2 = up^2 + wr^2.$$

Squaring the first and comparing with $(u+w)$ times the second gives $(up+wr)^2 = (u+w)\left(up^2 + wr^2\right)$, which simplifies to $uw(p - r)^2 = 0$. As $u, w \neq 0$, we get $p = r$, and then $q = p$ too. So all three columns share a single ratio s, and the grid is the top row multiplied by 1, s and s^2.

Now $a = 2$, $(2 + d)s = 9$ and $(2 + 2d)s^2 = 36$. Eliminating s gives $81(2 + 2d) = 36(2 + d)^2$, that is $2d^2 - d - 1 = 0$, so $d = 1$ or $d = -\frac{1}{2}$, with $s = 3$ or $s = 6$. Both grids

$$\begin{array}{ccc} 2 & 3 & 4 \\ 6 & 9 & 12 \\ 18 & 27 & 36 \end{array} \qquad\qquad \begin{array}{ccc} 2 & \frac{3}{2} & 1 \\ 12 & 9 & 6 \\ 72 & 54 & 36 \end{array}$$

satisfy every condition, and the bottom left entry $2s^2$ is 18 or 72. The sum is 90.

Three entries look too few to fix the five unknowns a, d, p, q, r, which is what the last option plays on; the hidden fact that the column ratios must be equal removes the freedom. Stopping at the integer solution gives 18.

Question 23. Answer **E**.

The tangent at (t, t^2) is $y = 2tx - t^2$. It passes through $P(p, p - 2)$ when $t^2 - 2pt + p - 2 = 0$. This always has two roots $a < b$, since its discriminant is $4\left(p^2 - p + 2\right) > 0$, and $b - a = 2\sqrt{p^2 - p + 2}$. The two tangents meet where $2ax - a^2 = 2bx - b^2$, that is at $x = \frac{a+b}{2} = p$.

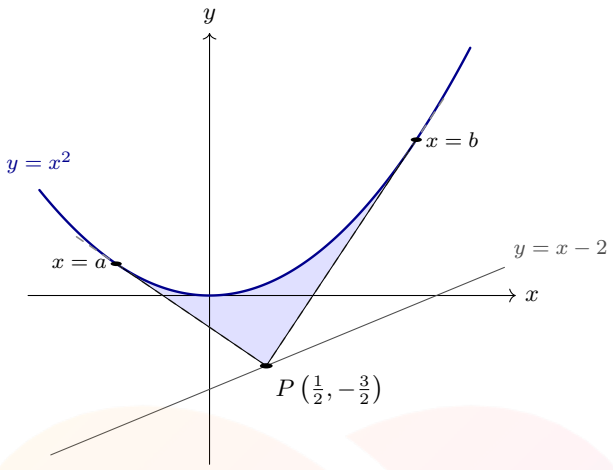
The region lies between the parabola and the tangent at a for $a \le x \le p$, and between the parabola and the tangent at b for $p \le x \le b$. Since $x^2 - \left(2ax - a^2\right) = (x - a)^2$, its area is

$$\int_a^p (x - a)^2\, dx + \int_p^b (x - b)^2\, dx = \frac{(p - a)^3 + (b - p)^3}{3} = \frac{(b - a)^3}{12}.$$

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So the area is $\frac{8}{12}\left(p^2-p+2\right)^{3/2} = \frac{2}{3}\left(\left(p-\frac{1}{2}\right)^2+\frac{7}{4}\right)^{3/2}$, which is least when $p=\frac{1}{2}$:

$$\frac{2}{3}\left(\frac{7}{4}\right)^{3/2} = \frac{2}{3}\cdot\frac{7\sqrt{7}}{8} = \frac{7\sqrt{7}}{12} \approx 1.54.$$



The region between the chord of contact and the parabola has area $\frac{(b-a)^3}{6}$, which gives $\frac{7\sqrt{7}}{6}$, and the whole triangle formed by P and the two points of contact gives $\frac{7\sqrt{7}}{4}$. Assuming by symmetry that P should be at $p=0$ gives $\frac{4\sqrt{2}}{3}$.

Question 24. Answer **A**.

Taking logarithms to base 2, the condition says that nx is within $\frac{1}{40}=0.025$ of an integer, where $x=\log_2\frac{3}{2}=0.58496$. Working out sixty multiples is slow. The key is that x is very close to $\frac{7}{12}=0.58333\ldots$ (which is why a piano has 12 notes to the octave). Write $x=\frac{7}{12}+\delta$, where $\delta=0.00163$ to 3 significant figures, so that

$$nx = \frac{7n}{12}+n\delta, \qquad 0<n\delta\le 60\delta<0.098.$$

If $\frac{7n}{12}$ is $\frac{j}{12}$ more than an integer with $1\le j\le 10$, then nx is at least $\frac{1}{12}$ above one integer and at least $\frac{2}{12}-0.098>0.06$ below the next, so it fails. If $\frac{7n}{12}$ is an integer, then 12 divides n and the distance is $n\delta$, which is less than 0.025 only for $n=12$ (it is 0.0195 there, but already 0.039 for $n=24$). If $\frac{7n}{12}$ is $\frac{1}{12}$ less than an integer, then $7n$ leaves remainder 11 on division by 12, so $n=5,17,29,41,53$, and the distance is $\left|\frac{1}{12}-n\delta\right|$, which is less than 0.025 only when $n\delta$ is between 0.058 and 0.108. This picks out $n=41$ (distance about 0.017) and $n=53$ (distance about 0.003), while $n=29$ misses by 0.036.

So the list is $12, 41, 53$. The 12-note scale is on it but its multiples are not, because the small error grows with each repetition.

Question 25. Answer **C**.

We must solve $y=3x-x^3$ and $x=3y-y^3$ together. Subtracting the equations gives $y-x=3(x-y)-\left(x^3-y^3\right)$ and adding them gives $x+y=3(x+y)-\left(x^3+y^3\right)$, so the pair is equivalent to

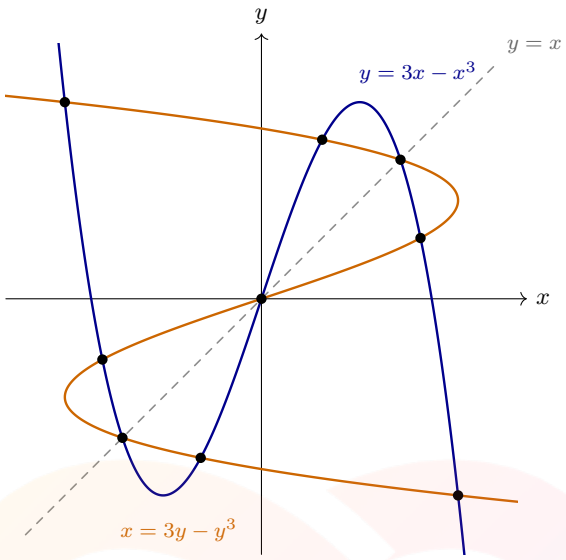
$$(y-x)\left(4-x^2-xy-y^2\right)=0 \qquad \text{and} \qquad (x+y)\left(2-x^2+xy-y^2\right)=0.$$

Now combine one factor from each.

$y=x$ with $x+y=0$ gives the origin. $y=x$ with $x^2-xy+y^2=2$ gives $x^2=2$, the points $\pm\left(\sqrt{2},\sqrt{2}\right)$. $x^2+xy+y^2=4$ with $y=-x$ gives $x^2=4$, the points $\pm(2,-2)$. Finally $x^2+xy+y^2=4$

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with $x^2 - xy + y^2 = 2$ gives $xy = 1$ and $x^2 + y^2 = 3$, so $(x + y)^2 = 5$ and $(x - y)^2 = 1$, which gives four more points, $\pm\left(\frac{\sqrt{5}+1}{2}, \frac{\sqrt{5}-1}{2}\right)$ and $\pm\left(\frac{\sqrt{5}-1}{2}, \frac{\sqrt{5}+1}{2}\right)$. All nine points are different, so the curves meet 9 times.



For an increasing function the curve and its reflection can meet only on the line $y = x$, and that reasoning gives the tempting answer 3. Here the curve is not monotonic, and six of the meeting points lie off the line $y = x$. Spotting the pair $\pm(2, -2)$ but not the other four gives 5.

Question 26. Answer **C**.

In coordinates, $P' = \dfrac{16}{x^2 + y^2}(x, y)$. Applying the map twice returns every point to where it started, so also $P = \dfrac{16}{x'^2 + y'^2}(x', y')$, where $P' = (x', y')$. Substituting, the condition $x > 4$ becomes $\dfrac{16x'}{x'^2 + y'^2} > 4$, that is

$$(x' - 2)^2 + y'^2 < 4.$$

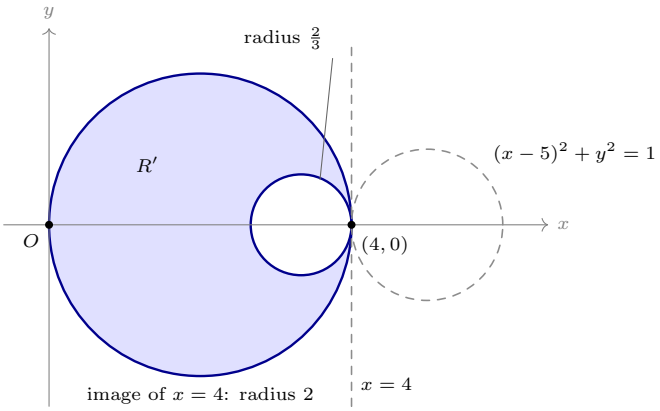
So the half-plane $x > 4$ is sent onto the disc of radius 2 centred at $(2, 0)$, with O itself left out. The far-away points of the half-plane crowd in towards O, which is how an unbounded region can have an image of finite area.

Similarly, the disc $(x-5)^2+y^2 < 1$, that is $x^2+y^2-10x+24 < 0$, becomes $\dfrac{256}{x'^2 + y'^2} - \dfrac{160x'}{x'^2 + y'^2} + 24 < 0$, that is

$$\left(x' - \tfrac{10}{3}\right)^2 + y'^2 < \tfrac{4}{9},$$

a disc of radius $\frac{2}{3}$. As a check, the ends $(4, 0)$ and $(6, 0)$ of a diameter go to $(4, 0)$ and $\left(\frac{8}{3}, 0\right)$, the ends of a diameter of the image. Note that the centre $(5, 0)$ does not go to the centre of the image. This small disc lies inside the big one, touching it at $(4, 0)$.

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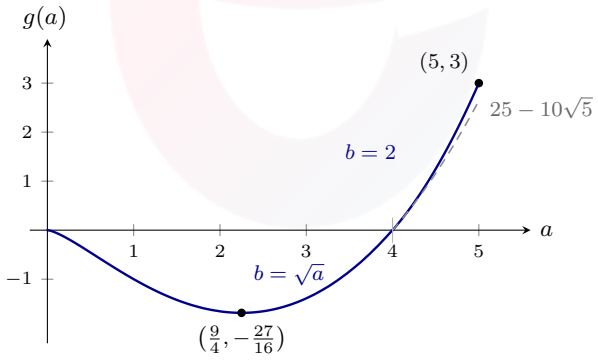
So R' is the big disc with the small disc removed, and its area is $4\pi - \frac{4\pi}{9} = \frac{32\pi}{9}$. Forgetting to remove the small disc gives 4π, assuming the small circle is unchanged gives 3π, and adding instead of subtracting gives $\frac{40\pi}{9}$.

Question 27. Answer **A**.

For a fixed a, Bob minimises $P(b) = b^3 - 3ab + a^2$ over $0 \le b \le 2$. Here $P'(b) = 3b^2 - 3a$, so P decreases until $b = \sqrt{a}$ and increases after it.

If $a \le 4$, then $\sqrt{a} \le 2$ and Bob takes $b = \sqrt{a}$, paying $g(a) = a^2 - 2a^{3/2}$. If $4 < a \le 5$, then P is decreasing on the whole of $0 \le b \le 2$, so Bob takes $b = 2$ and pays $g(a) = a^2 - 6a + 8$.

Alice now maximises g. On $0 \le a \le 4$, $g'(a) = 2a - 3a^{1/2}$ vanishes at $a = \frac{9}{4}$, but this is a minimum, with $g = -\frac{27}{16}$; the largest values on this piece are $g(0) = g(4) = 0$. On $4 \le a \le 5$, $g(a) = (a-3)^2 - 1$ increases from 0 to $g(5) = 3$. So Alice picks $a = 5$, Bob replies with $b = 2$, and Bob pays $8 - 30 + 25 = 3$ pounds.



Forgetting that Bob cannot go beyond $b = 2$ gives $g(5) = 25 - 10\sqrt{5} \approx 2.64$; stopping at the stationary point gives $-\frac{27}{16}$; looking only at $a \le 4$ gives 0; and 8 and 25 come from fixing $b = 2$ or $b = 0$ for every a.

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